Estimating NHL Screen Assists from Public Replay Tracking

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Abstract

We estimate offensive screen assists from public NHL goal-replay tracking and evaluate player attribution against independent Clear Sight Analytics (CSA) annotations. The study covers 2023–24 through 2025–26 and completed 2026–27 games through October 19, 2026. We compare a geometric baseline with ridge logistic regression, separating development, model selection, and evaluation by season. On the reserved 2025–26 reference set, conservative attribution achieves 92.60% precision and 73.21% recall, while a broader definition achieves 84.42% precision and 85.85% recall. The resulting player rankings include counts, rates based on official appearances and ice time, and a replay for each credited goal. In a separate mixed-effects analysis of reviewed CSA shots, standardized scoring rates are 6.15% with offensive screens and 2.07% with clear sight. Together, these analyses provide an evaluated method for describing screening contributions on goals and evidence of screening's association with scoring. Interpretation remains limited by planar tracking, uncertain shot timing, selected reference labels, and sensitivity to positional assumptions; the estimates do not identify causal goals added by individual players.

Keywords: ice hockey, player attribution, tracking data, screening, mixed-effects logistic regression

1 Introduction

An offensive teammate who obstructs the goaltender's view can contribute to a goal without receiving an official assist. A screen-assist statistic can record that contribution, provided its definition accounts for the timing and location of the obstruction. A player near the crease may leave the puck visible, while a player crossing the sightline briefly may interfere with the goalie's view around release. We therefore examine the scoring shot and the movement of potential screeners over a short interval, then evaluate whether the resulting credits identify independently annotated participants.

Earlier work provides several approaches to measuring screening and traffic. Cait (2019) manually identifies screen assists on goals, including stationary and moving screens. Using tracking data, Pitassi et al. (2025) examines traffic in the area between the puck and goalposts and its relationship with shot outcomes, accounting for differences in shot location. Their analysis includes whether attempts reach the net. In related work, Iaboni et al. (2025) combines tracking with video-derived player poses to estimate the visibility and reachability of the net from the puck's perspective. These studies address related but distinct quantities: players in a shooting lane, the area of net available to a shot, and obstruction of the goalie's view need not coincide.

We use public NHL goal replays to identify offensive teammates whose tracked positions resemble independently annotated screens. Clear Sight Analytics (CSA) provides the private reference for development and evaluation, while official appearances and ice time supply the exposure denominators for player rankings. Readers can inspect each credited goal through an accompanying replay explorer. To examine the relevance of screening beyond goals alone, we also analyze its association with scoring in a separate CSA shot population. This distinction between attribution on goals and scoring outcomes is central to interpreting the rankings.

2 Data and study populations

2.1 Public goals and player exposure

The snapshot includes completed regular-season and playoff games in 2023–24, 2024–25, and 2025–26, plus 2026–27 games dated through October 19, 2026. We exclude shootout events and acquire public data with `nhlscraper` version 0.7.0.9000 (Saijo et al. 2026), recording the package revision and public database snapshot in the repository. Official skater appearances and time on ice provide the exposure denominators, with team membership assigned separately for each game.

Public replays contain planar puck and player coordinates, identities, and timestamps at an observed sampling interval of 0.10 s. Sequences generally span about 12–14 s, although their endpoints do not define shot release or goal time. We sort the timestamps, retain the first observation at duplicated timestamps, and follow player identifiers across tracking slots. Coordinates are expressed in feet on a rink centered at the origin.

The public population contains 26,457 non-shootout goals. Stored replay availability in 2023–24 is 99.22% before tracking-quality exclusions; across all included seasons, 22,570

Table 1: Public goal population after shootout exclusion. The 2026–27 sample includes completed games through October 19, 2026.

Season	Goals	Evaluated	Empty net	Unevaluable
2023–24	8,590	7,379	485	726
2024–25	8,435	7,098	568	769
2025–26	8,572	7,336	547	689
2026–27	860	757	44	59

goals have a usable shot interval and candidate teammates. Table 1 distinguishes these evaluated goals from empty-net goals and other unevaluable sequences. We retain an unevaluable status when the tracking cannot support attribution.

2.2 Independent reference annotations

We query CSA's historical reporting database read-only and retain private records in memory. Reference events must be game-reviewed, quality-control complete, classified as shots, and associated with an identifiable goalie. Matches require agreement on game identity, scorer, shooting team, home and away teams, goalie, and period. An imported event-identifier match allows a clock difference of at most one second. When no valid identifier match is available, we require a unique exact clock-and-scorer match with the same contextual checks. Participants are mapped through NHL identifiers, and duplicate event–player–side memberships are removed.

Positive events require an explicit offensive-screen, mixed-screen, one-timer/screen, or screen/deflection classification and at least one named offensive teammate other than the scorer. Negative events require an explicit clear-sight or defensive-only classification with no offensive teammate recorded. We exclude screen-on-pass cases and screen-classified events lacking a named offensive teammate, so absence of a participant row alone never establishes a negative event.

Within an event with recorded participants, their list supplies the player-attribution reference. We treat other candidate teammates as negatives and count named participants missing from usable tracking as unrecovered positives. This assumes the participant list is complete. Because the available documentation also lacks an explicit operational threshold for brief screens or frame-level screen timing, disagreement may reflect differences in coding scope as well as detector error. Table 2 describes the matched and evaluable populations; their screening prevalence follows the eligibility rules and need not match the prevalence among all NHL goals.

3 Estimating a screen assist

3.1 Release and arrival

We rotate each sequence so the defending goal is at $x = 89$ ft and identify goal-plane crossings toward the net. A crossing qualifies when its interpolated lateral coordinate lies

Table 2: CSA reference populations. Matched goals precede label and replay-quality restrictions.

Season	Matched goals	Benchmark goals	Screen-positive goals
2023–24	7,587	1,687	511
2024–25	7,245	1,730	528
2025–26	7,813	1,747	494

within 4.50 ft of the goal center, allowing some tracking displacement around the 6 ft goalmouth. Arrival is assigned to the first sample at or beyond the plane. We then search backward up to 2.00 s for a release, using puck speed and agreement with the official shot location to assess candidate intervals.

For ordinary shots, a release candidate moves toward goal at at least 12 ft/s, lies within 22 ft of the recorded shot location and 12 ft of the scorer's sensor, and is followed by at least 2 ft of additional puck–scorer separation over the next three samples, or fewer if arrival occurs first. Three samples normally span 0.30 s. We choose the candidate minimizing location error divided by 10 ft plus scorer separation divided by 5 ft. These tolerances accommodate a stick release displaced from the shoulder and coarse official shot locations, without directly observing contact with the stick.

For tips and deflections, we trace the incoming flight backward through consecutive fast-moving samples and check the credited scorer near the terminal deflection to approximate the original release. Rebound goals use the scoring attempt and its recorded location; earlier traffic contributes only if it also falls within that attempt's analysis window. We require at least 0.50 s of preceding tracking and reject critical intervals containing gaps longer than 0.21 s or missing puck or goalie positions. Two candidate arrivals more than 0.50 s apart with matching costs differing by less than 0.25 make the sequence ambiguous. These rules reconstruct shot timing from the available trajectories, with accuracy limited by 10 Hz sampling, planar coordinates, and the absence of puck-touch annotations.

3.2 Finite sightline features

At time t, let g_t, p_t, and s_{it} be the tracked goalie, puck, and candidate player's positions. With $v_t = p_t - g_t$, define

$$u_{it} = \frac{(s_{it} - g_t)^\mathsf{T} v_t}{v_t^\mathsf{T} v_t}, \qquad d_{it} = \left\| (s_{it} - g_t) - u_{it} v_t \right\|.$$

We evaluate obstruction when $0 < u_{it} < 1$ and the segment exceeds 2 ft, restricting attention to players between the goalie and puck. The feature windows cover the 0.50 s before release, a band from -0.20 to $+0.10$ s around release, and flight toward the goalie. Post-release windows stop at arrival and exclude puck positions more than 1 ft beyond the goalie's longitudinal position. Duration calculations weight intervals centered on observed samples by their overlap with the relevant window. Reacquisition time is the interval since the most recent obstruction in the preceding 1.00 s, capped at 1.00 s when none is observed.

Candidate players require at least three valid samples and 80% observed positions in the feature window.

The 16 predictors describe minimum and release-time separation, obstruction duration under primary and narrower/wider tolerances, longest continuous duration, flight obstruction, alignment strength before/near/after release, reacquisition time, goalie proximity, relative depth, crossing, shot distance, and flight time. For an effective radius r, alignment strength is

$$A_{it}(r) = \frac{\exp\{-d_{it}^2/(2r^2)\}}{1 + \|s_{it} - g_t\|/20},$$

with zero strength outside the finite segment. The geometric baseline combines peak near-release or flight alignment, pre-release obstruction duration D_i, and reacquisition time R_i, both measured in seconds:

$$B_i = \max(A_{i,\text{near}}, A_{i,\text{flight}})\{0.25 + \min(1, D_i/0.30)\}\exp(-R_i/0.40).$$

This score gives greater weight to close alignment, sustained obstruction, and a shorter interval for reacquisition before release.

3.3 Position and timing uncertainty

The right-shoulder sensor placement described by Radke et al. (2021) displaces tracked positions from the player's body center and, for a goalie, from the eyes. Movement direction does not resolve orientation because players can skate backward, move sideways, or remain stationary. Public replays also lack puck height, gaze, stance, and the extent to which limbs or equipment obscure vision. The finite segment consequently serves as a planar proxy for a three-dimensional visual process.

We use an effective tolerance of 1.50 ft and examine alternatives of 1.00 and 2.00 ft to assess sensitivity to body extent and sensor displacement. We separately shift estimated release by -0.10 and $+0.10$ s while retaining the selected model and thresholds. If a shifted feature window lacks sufficient tracking, its primary feature values are retained. These bounded scenarios describe dependence on positional and timing assumptions; they are not calibrated distributions of anatomical error and do not cover every possible posture.

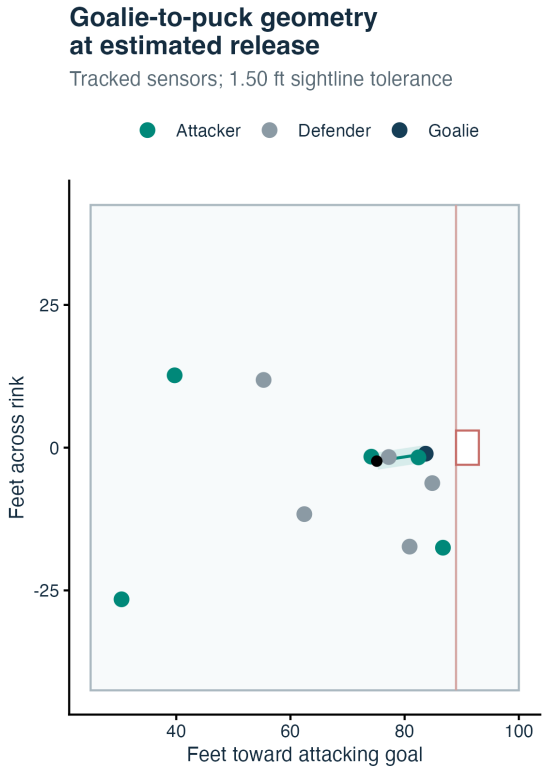


Figure 1: An actual replay at estimated release. Tracked sensor positions define the finite goalie-to-puck segment; the band illustrates effective geometric tolerance.

3.4 Model selection and attribution

We develop on 2023–24, select the model and thresholds on 2024–25, and reserve 2025–26 for final evaluation. Splitting by season keeps entire games together. We fit ridge logistic regression through `tidymodels` and `glmnet`, using standardized public features and an unpenalized intercept; player identities and CSA scoring probabilities are excluded from the predictors. We compare the geometric baseline with five requested ridge penalties, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, and 1, along the fitted regularization path. Requests below the path's minimum use its terminal coefficients. Selection uses player-level average precision, grouping tied scores and preferring the baseline in a tie.

For the selected detector, the conservative threshold maximizes

$$F_{0.5} = \frac{1.25\,PR}{0.25P + R},$$

where P and R denote precision and recall. The broader threshold maximizes $F_1 = 2PR/(P+R)$ subject to being no higher than the conservative threshold. The selected values are 0.6155 and 0.2443, respectively, and we freeze coefficients and thresholds before final-season evaluation. Each qualifying offensive teammate receives one credit per goal, excluding the scorer. Multiple teammates can qualify, including official assist recipients,

and every conservative credit is included in the broader total. A goal-level prediction is positive when at least one teammate qualifies.

We report precision, recall, average precision, and, for the classifier, calibration. Threshold-specific 95% intervals use a game-clustered sandwich calculation for each ratio. For example, with precision $\hat{P} = \sum_g T_g / \sum_g N_g$, the estimated variance is

$$\widehat{\mathrm{Var}}(\hat{P}) = \frac{G}{G-1} \frac{\sum_g (T_g - \hat{P} N_g)^2}{(\sum_g N_g)^2}.$$

Here G is the number of games represented in the benchmark. We truncate Wald interval endpoints to $[0, 1]$; the intervals describe benchmark sampling variation conditional on the selected detector and reference labels. Annotation uncertainty and model-selection uncertainty are not included.

To examine possible failure mechanisms, we review tracking frames for 12 development-season player–goal cases across 11 goals, plus an additional unresolved trajectory. The cases include strong alignment, participant disagreements, multiple potential screeners, tips, a rapid follow-up scoring attempt, and a crossing near an attribution threshold. Comparing frames before, at, and after estimated release helps interpret the numerical features, although tracking-frame review cannot independently establish what the goalie sees and provides no additional accuracy estimate.

3.5 Exposure and rankings

For player i under a common set of season, competition, team, and cutoff filters, we compute

$$\text{Screen assists per game}_i = \frac{C_i}{G_i},$$
$$\text{Screen assists per 60 min}_i = \frac{3600 C_i}{T_i},$$

where C_i is estimated credits under the selected definition, G_i is official appearances, and T_i is ice time in seconds. Traded players' records follow their team in each game, and players with no detected credits remain eligible. Count rankings impose no minimum. For rate rankings, completed seasons and the combined all-seasons view default to 25 games or 250 minutes; the ongoing season defaults to five games or 60 minutes. Playoff-only rankings start without a minimum. Readers can adjust these thresholds, and rates always require positive exposure.

The denominators include all official exposure, whereas credits come from evaluable goal replays. The rates therefore describe detected screen assists per game or unit of playing time, including exposure from games with unresolved tracking. Missing replays can depress these rates, and we do not impute additional credits for unevaluable goals.

4 Screening and scoring

We examine scoring in a separate CSA population from 2024–25 and 2025–26, when release-type information is available. CSA describes its central coverage as shot sequences resulting in shots on goal (Clear Sight Analytics n.d.), so the database should not be treated as a complete record of blocked and missed attempts. We compare reviewed, quality-complete offensive-screen and clear-sight shots with a goalie present, excluding screen-on-pass cases, posts, and deflection releases. Mixed screens, defensive screens, one-timer/screen categories, and rebound categories are outside this primary contrast. To restrict comparison to shared observed contexts, each included season–strength–precise-location–release stratum must contain both screened and clear-sight observations.

The prespecified mixed-effects logistic model is

$$\operatorname{logit}\{\Pr(Y_j = 1)\} = \alpha + \beta S_j + x_j^{\mathsf{T}}\gamma + b_{\text{shooter}(j)} + b_{\text{goalie}(j)} + b_{\text{game}(j)}.$$

Here Y_j indicates a goal and S_j indicates an offensive screen. The covariate vector x_j contains season, regular-season/playoff status, strength, precise shot zone, release type, broad play context, first and second preceding-pass zones, and attacking and defending teams. Shooter, goalie, and game intercepts follow independent mean-zero normal distributions, and missing pass or context values are retained as an explicit unknown category. CSA scoring probabilities are excluded from the model.

We fit this model through `tidymodels` with the `multilevelmod/lme4` engine, using penalized iteratively reweighted least squares (`nAGQ = 0`). For standardization, we predict screened and clear-sight outcomes over the covariate distribution of screened shots in the shared strata, integrating the fitted random-intercept distributions with 41 deterministic normal-quantile nodes. The adjusted odds ratio is the conditional contrast $\exp(\beta)$, with a fixed-effect Wald interval. Delta-method intervals describe the standardized rates and their difference. All interval calculations hold variance components at their fitted estimates.

5 Results

5.1 Independent attribution performance

The selected ridge detector has tuning-season average precision 0.893, compared with 0.835 for the geometric baseline. On the reserved season, player-level average precision is 0.911 and goal-level average precision is 0.931. This final benchmark contains 1,747 goals and 7,260 player–goal units, including 530 annotated offensive teammate contributions. Conservative attribution recovers 388 contributions and assigns 31 credits to participants absent from the reference. The broader definition recovers 455 contributions, with 84 such disagreements.

Goal-level detection has higher precision and recall than participant attribution under both definitions (Table 3), illustrating the distinction between detecting a screened goal and identifying its participants. Calibration adds a further qualification: the held-out Brier score is 0.0171, while intermediate-score bins have higher observed participant frequencies than their mean predictions. The large concentration of nonparticipants in the lowest

Table 3: Reserved 2025–26 performance, with game-clustered 95% intervals.

Definition	Task	Precision (%)	Recall (%)
Conservative	Player	92.60 (90.04, 95.16)	73.21 (69.56, 76.85)
Conservative	Goal	93.70 (91.31, 96.09)	75.30 (71.51, 79.10)
Broader	Player	84.42 (81.41, 87.42)	85.85 (82.92, 88.78)
Broader	Goal	87.27 (84.35, 90.19)	87.45 (84.57, 90.33)

bin contributes to the low overall score. We use classifier scores with empirically selected attribution thresholds; their calibration outside the selected reference population remains unknown.

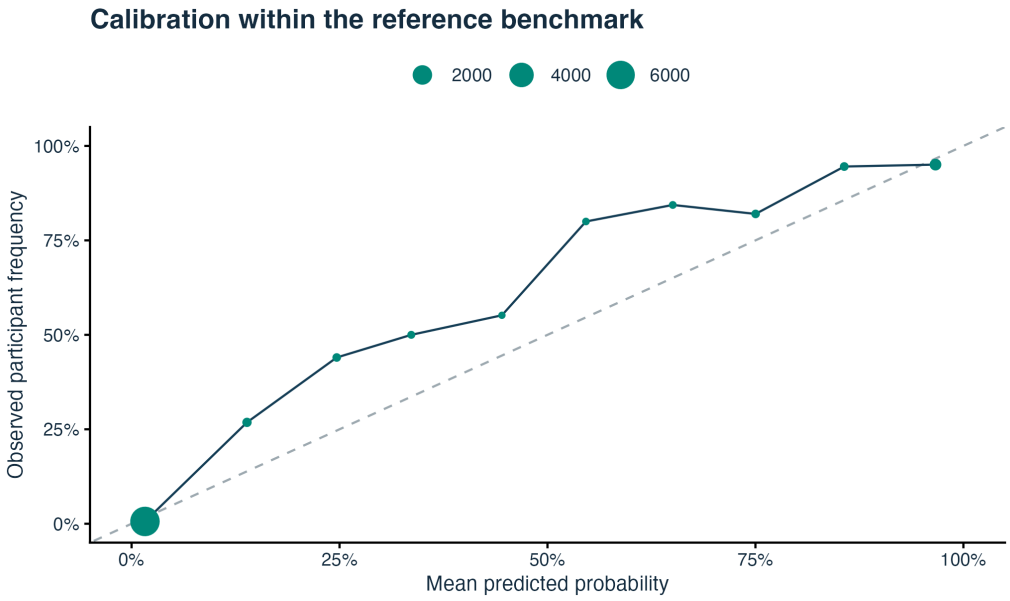


Figure 2: Calibration on the reserved reference set. Point size reflects bin count. Intermediate-score bins show underprediction, and calibration outside this selected benchmark is unknown.

5.2 Player rankings and sensitivity

Across the included seasons and competitions, the detector assigns 2,211 conservative and 3,320 broader credits. In the 2025–26 regular season, Alexander Wennberg leads with 13 conservative credits; Joel Eriksson Ek and Chris Kreider each receive 10. Table 4 reports both totals and rates for the leading players. The website opens on 2025–26, the most recent completed season, and connects every credited goal to its replay.

Table 4: Leading conservative screen-assist totals, 2025–26 regular season. Rates use conservative credits.

Player	Conservative	Broader	GP	Screen assists per game	Screen assists per 60 min
Alexander Wennberg	13	14	80	0.16	0.48
Joel Eriksson Ek	10	15	70	0.14	0.45
Chris Kreider	10	11	75	0.13	0.47
Anze Kopitar	9	9	67	0.13	0.42
Alex Tuch	7	9	79	0.09	0.28
Corey Perry	7	8	72	0.10	0.43
Jake Guentzel	7	8	81	0.09	0.26
Anders Lee	6	10	82	0.07	0.28
Charlie Coyle	6	9	82	0.07	0.24
Dmitri Voronkov	6	8	63	0.10	0.41

A zero indicates that no contribution is detected under the selected definition and filters, and may occur for a player who screens on other shots. Inspection of the development examples also shows how quickly alignment changes: a player can cross the segment within a few samples, while a named participant can remain outside the sensor-defined segment with body position unobserved. In the unresolved example, the tracked puck crosses the goal plane 6.38 ft from its center, beyond the goalmouth tolerance, so we retain an unevaluable status.

Both effective tolerance and estimated release timing change the number of qualifying contributions (Table 5). These differences demonstrate that the leaderboard depends on measurement assumptions as well as the selected attribution threshold. Conservative and broader totals describe two operating definitions, while the sensitivity scenarios examine additional positional and timing uncertainty.

Table 5: 2025–26 regular season and playoffs combined: fixed-detector sensitivity to geometry and release timing.

Scenario	Conservative	Broader
Primary: 1.50 ft	709	1,077
Tolerance: 1.00 ft	535	865
Tolerance: 2.00 ft	863	1,276
Release: 0.10 s earlier	800	1,209
Release: 0.10 s later	581	961

5.3 Adjusted scoring association

The common-support population contains 3,697 screened shots and 54,690 clear-sight shots across 2,792 games, retaining 96.40% of otherwise eligible screened shots. Observed screened

scoring rates are 5.86% in 2024–25 and 6.45% in 2025–26; the corresponding clear-sight rates are 3.10% and 3.20%.

After adjustment and standardization to screened-shot covariates, the screened scoring rate is 6.15 (5.43, 6.88) percent and the clear-sight rate is 2.07 (1.88, 2.25) percent, with 95% intervals in parentheses. The standardized difference is 4.09 (3.34, 4.84) percentage points, and the conditional adjusted odds ratio is 3.47 (2.92, 4.13). These contrasts describe screening's association with scoring within the included shot population.

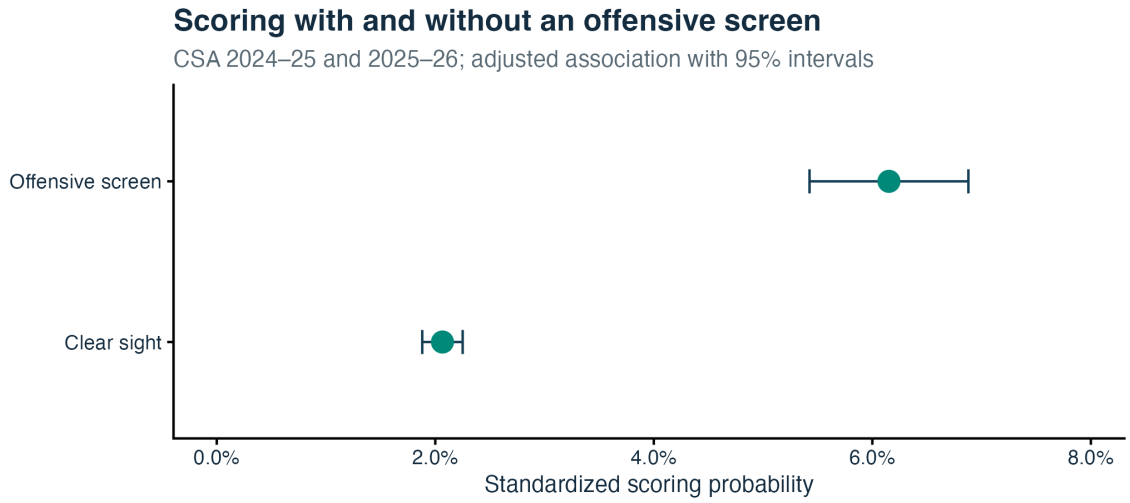


Figure 3: Standardized scoring rates with model-based 95% intervals. Both predictions use screened-shot covariates within common support.

The optimizer converges with a goalie random-intercept variance of zero. Shooter and game variances are approximately 0.0052 and 0.0049 on the log-odds scale. We retain the boundary fit, with interval calculations conditional on these variance estimates. Unmeasured shot characteristics, detailed pass timing, player selection, and classification practices remain possible contributors to the association.

6 Discussion

Public replay tracking identifies many independently annotated screen participants under a relatively simple detector. The conservative threshold prioritizes precision, while the broader threshold recovers more reference contributions at the cost of additional disagreements. Presenting both totals alongside the credited plays makes that tradeoff available for inspection and gives the rankings a defined empirical basis.

The rankings nevertheless condition on a goal being scored. Screens on shots that are saved, miss the net, or are blocked cannot receive credit in this dataset, and team shooting volume, offensive role, special-teams deployment, and teammate finishing affect opportunities. Official appearances and ice time provide consistent exposure denominators, but they do not remove these differences. The counts and rates consequently describe detected contributions on goals rather than a player's complete screening activity.

Measurement uncertainty is also substantial. The goalie-to-puck segment approximates a three-dimensional visual process using shoulder-mounted positions, and brief crossings can occur between samples. Deflections, rebounds, and goalie contact complicate release inference, while unevaluable trajectories may be distributed unevenly across teams or types of play. Sensitivity to effective tolerance and release timing shows that apparent differences between players can depend on assumptions about unobserved positions and events.

Reference construction introduces further limits. Explicit classifications and participant identities support evaluation, but the eligibility rules leave ambiguous plays outside the benchmark and change screening prevalence. Named participants can be misclassified, and unlisted teammates may still obstruct vision. The reported performance therefore measures agreement with a defined reference population. Validation on a broader set of independently reviewed plays, ideally combining tracking with video and clearer annotation rules, would help establish how the method performs beyond that population.

The CSA scoring analysis provides complementary evidence that offensive screens are associated with scoring after adjustment for observed context. Its population and estimand differ from those of the goal-replay detector, however, so neither the adjusted odds ratio nor the standardized rate difference can be multiplied by a player's screen-assist total to estimate causal goals added. That question requires broader tracking, an account of shot selection, and a defensible counterfactual design. The present rankings provide a reproducible description of attributed screens on goals, with replay inspection and measured reference agreement available to support their use.

7 Data and code availability

The website provides rankings, replay exploration, methodology, and this paper. The repository contains R code, editable Quarto source, explicit detector coefficients, NHL-derived goal–player credits, player-game exposure, short replay segments, and aggregate findings. Provenance records the `nhlscraper` revision and public database snapshot. These artifacts reproduce the published ranking arithmetic, figures, and report. CSA records, event-level labels, and fitted private model objects are not distributed; re-estimating the CSA-dependent models requires authorized access. The publication is a fixed snapshot through October 19, 2026.

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